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Fostering Business Growth Through Local Content Programs: Case Study from Kingdom of Saudi Arabia

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Abstract

This study explores the transformative role of local content programs in fostering business growth and driving economic diversification in the Kingdom of Saudi Arabia. These programs, essential components of the 2030 vision, aim to promote the use of local goods and services, workforce development, and technology transfer. By assessing their mechanisms and impact, the research provides strategic insights into how these programs influence supply chain management and investment attractiveness in Saudi Arabia's oil and gas sector.

The research workflows show the role of local content programs in promoting business growth through organizations. The methodology investigates the alignment of local content programs with Saudi Arabia's Vision 2030 goals, focusing on partnerships with key stakeholders. The study also examines components such as local goods and services, labor compensation, training and development, supplier growth, and research investments. By unifying local content metrics and certifications, the research underscores how standardization efforts enhance supply chain reliability and promote industrial resilience. The study provides insights from market intelligence tools and case studies to evaluate the design and execution of local content initiatives.

The findings underscore the strategic importance of local content programs as a catalyst for business and economic growth. Key results include:

- Enhanced operational frameworks: Local content programs have streamlined procurement processes and created robust frameworks for integrating local goods, services, and talent into business operations.
- Increased economic impact: Companies achieved higher localization rates in their supply chains, fostering job creation, workforce skill development, and technology transfer while supporting economic diversification.
- Challenges and strategic responses: Varied program requirements across sectors introduced operational complexities, prompting companies to develop specialized administrative units to monitor and optimize compliance.
- Quantifiable business value: Organizations reported measurable cost savings, reduced reliance on imports, and improved operational efficiencies, further aligning with Vision 2030's strategic goals.

The study concludes that harmonizing local content programs and fostering public-private partnerships can further enhance their effectiveness. Standardized policies will simplify compliance, reduce administrative burdens, and make the region more attractive to investors.

This case study provides a unique lens into the evolution of Saudi Arabia's local content programs, emphasizing their dual impact on business strategy and economic development. By bridging operational realities with strategic goals, the findings offer a roadmap for aligning corporate objectives with national economic policies. This research contributes to the broader discourse on supply chain localization and the interplay between regulatory frameworks and business growth in resource-rich economies.

Key Words: Local Content Programs, Vision 2030, Economic Diversification, Business Growth, Supply Chain Localization, Saudi Arabia, Oil and Gas Sector, Investment Strategies

Introduction

The Kingdom of Saudi Arabia stands at a pivotal juncture in its economic history. Historically, the nation's economy has been heavily reliant on revenues generated from its vast oil reserves. However, with the global energy landscape undergoing a significant transformation and a growing emphasis on sustainable and renewable energy sources, the Kingdom recognizes the urgent need to diversify its economic base. This imperative has led to the formulation of vision 2030, an ambitious and comprehensive strategic framework designed to propel Saudi Arabia towards a more sustainable and diversified economic future ([Deringer et al., 2018](#); [Emielu, 2005](#)). Vision 2030 encompasses a wide array of strategic objectives, including but not limited to the development of non-oil sectors, fostering a vibrant private sector, attracting foreign direct investment, and creating a highly skilled and competitive workforce. At the heart of vision 2030 lies the recognition that long-term economic prosperity cannot solely depend on the fluctuating global oil market. Instead, it requires a concerted effort to build a robust and diversified economy capable of competing effectively in the global arena ([Acheampong et al., 2016](#)).

Local content programs are critical to realizing the ambitious goals of Saudi Arabia's vision 2030. These programs are designed to promote the utilization of locally manufactured goods, the engagement of local service providers, and the development of a skilled Saudi workforce. Through prioritizing local content, the Kingdom aims to stimulate domestic economic activity, create new job opportunities for its citizens, foster innovation and technological advancement, and reduce its reliance on foreign imports. In essence, local content programs are intended to serve as a catalyst for sustainable economic growth and diversification, aligning seamlessly with the overarching objectives of vision 2030 ([Alayed et al., 2025](#)). Local content policies have emerged as a prominent feature of economic development strategies in resource-rich countries worldwide. These policies are generally designed to increase the participation of local businesses and workers in the value chains associated with natural resource extraction and other major economic activities. The rationale behind local content policies is multifaceted, encompassing objectives such as promoting economic diversification, creating employment opportunities, fostering technological capabilities, and enhancing social inclusion ([d'Almeida et al., 2019](#); [Hammann, 2016](#); [Seregin, 2014](#)).

Several initiatives are underway in Saudi Arabia to standardize local content metrics. The in-kingdom total value added (IKTVA) program is a leading example of such an initiative. The IKTVA program covers a wide range of activities, including local procurement, workforce development, technology transfer, and investment in local facilities. Localized goods and services focus on leveraging local suppliers and contractors to provide essential goods and services, ensuring that spending remains within the country, thus contributing to local economic growth. Compensation paid to Saudis highlights the salaries and benefits given to Saudi nationals working in the oil and gas sector, emphasizing the program's goal of creating job opportunities for local citizens. The component of training & development of Saudis highlights the importance of skill-building and professional development initiatives aimed at enhancing the local workforce's capabilities. Additionally, supplier development involves initiatives designed to improve the

capacity and quality of domestic suppliers, ensuring they can effectively meet industry needs. Research and development element is another critical element, reflecting the program's encouragement for investments that foster innovation and strengthen local technology in the oil and gas sector. In relation to revenue emphasizes the significance of measuring and maximizing the economic returns from investments to the Kingdom (IKTVA, 2025).



Figure 1—IKTVA Elements (Zainaddin, 2022)

Local Content Requirements (LCRs) in the World Economy

Several major countries actively use local content requirements (LCRs) today, including Argentina, Brazil, China, India, Indonesia, Russia, Saudi Arabia, and the United States, as shown in Figure 2. Among these, India stands out as the world's largest user by a significant margin, employing over one-quarter of all global LCRs. Brazil also relies heavily on these rules as part of its recent economic policy, as reports show Brazil has introduced more LCRs than any other nation since 2008. These LCRs requirements cover a wide range of important industries such as technology, energy, healthcare, media, insurance, textiles, machinery, oil and gas, and finance (Deringer et al., 2018). For a broader view, Figure 3 lists the top 15 countries currently using LCRs worldwide and shows the key sectors where these rules are most commonly applied.

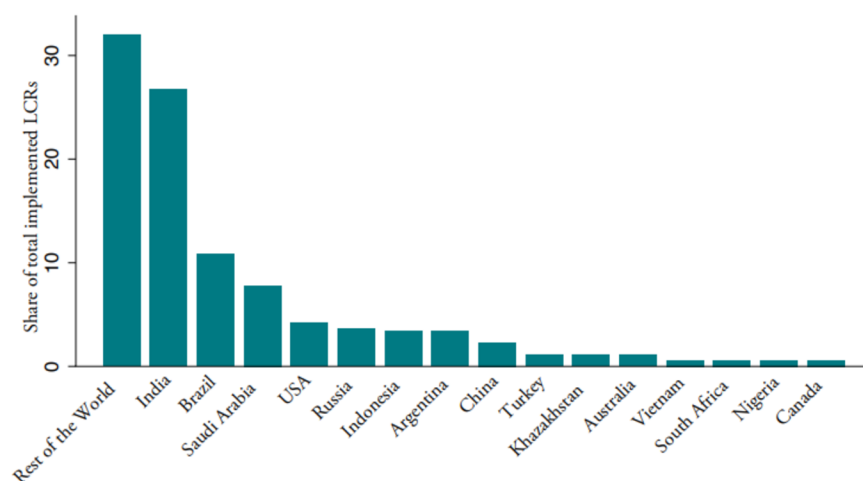


Figure 2—Global Distribution of LCR Implementation by Country (Deringer et al., 2018)

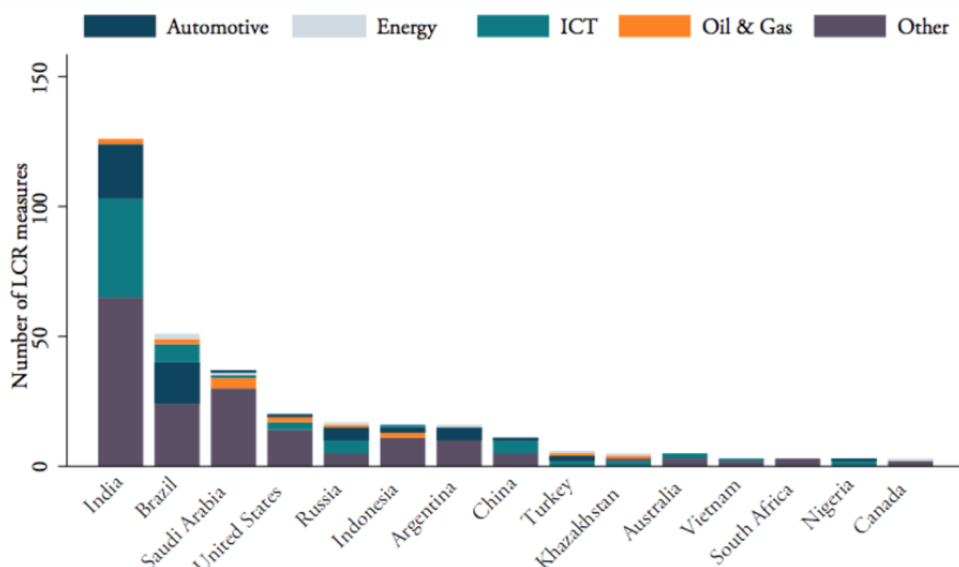


Figure 3—Implemented Local Content Requirements Overview (Deringer et al., 2018)

Saudi Arabia stands out as a significant implementer of local content requirements relative to other individual nations, with LCR implementation ranks among the top countries listed, with a strong focus on domestic value creation and economic diversification within Saudi Arabia's national strategies. Saudi Arabia's local content requirements are strategically distributed across multiple sectors. Figure 3 illustrates that Saudi Arabia's measures span beyond just Oil & Gas, incorporating information and communications technology (ICT) and other diverse industries with a strategic 2030 vision to have a comprehensive approach to promoting domestic value creation across its economy, rather than relying solely on its dominant energy sector.

While large, high-technology capital expenditures (CAPEX) investments are typically the most rigorously scrutinized contracting processes in terms of local content, the recurrent nature of OPEX presents a distinct and often underutilized opportunity for in-country value development. In-country value (ICV) refers to the total economic benefit captured and retained within the host country from industry activities, encompassing local procurement of goods and services, employment and development of the national workforce, and investment in local capabilities and infrastructure. Unlike the fluctuating nature of CAPEX investments, OPEX spending offers stability over time, making it particularly valuable for sustainable LC and ICV growth. This stable approach facilitates continuous LC development, which is crucial for building a skilled local workforce, and helps to mitigate the disruptive "stop and go" cycles inherent in the oil industry, a significant challenge for long-term development. Consequently, host countries should aim for a more balanced distribution of local content requirements between CAPEX and OPEX to fully optimize overall value creation and maximize in-country value retention (Hammann, 2016).

Executing the Methodology Workflow

The supply chain in the oil and gas sector (Figure 4) comprises several distinct activities or phases, beginning with exploration and concluding with distribution. Numerous tasks must be executed to ensure that the necessary goods and services are accessible at every stage of this process. Strategic decisions within the supply chain are crucial for the success of operations. Given the substantial resources required to carry out these tasks, this sector has been at the forefront of local content initiatives (Denney, 2011).

Goods	Exploration	Development	Production	Oil & Gas Treatment & LNG	Transport & Storage	Refining	Petrochem.	Primary Distribution
Subsea Equipment	Wellheads , Sub-surface Safety Valves, Compressors, Meters, Separators, Risers, Umbilicals							
Downhole Equipment	Casing Hardware, Completion Equipment , Drilling tools, Wireline Logging Tools, Perforating Systems							
Tubular Goods	Drill Pipe, Casing, Tubing, Manifolds							
Rigs, Platforms & FPSOs	Land Rigs, Offshore Fabrication, Vessel Conversions, Rig Equipment, Unit Manufacturing			Steel Structures, Production Topsides	Steel Structures			
Rotating Equipment				Compressors, Blowers, Turbines & Pumps				
Static Equipment				Surface Equipment, Columns & Exchangers	Transport Pipes, Tanks	Columns & Exchanges (e.g. Reactors, Vessels)		
Pipes, Valves & Fittings				Pipes, Valves & Fittings				
Electrical Equipment				Transformers & Switchboards, Drive Motors, Cables				
Instrumentation & Control	Sub-surface Sensors, Surface Production Monitoring (e.g., Separators, Multi-phase Flowmeters)			Control Systems & Valves, Instruments & Analysers				
Fluids & Chemicals	Drilling & Completion Fluids, Upstream Specialty Chemicals			Catalysts & Additives				
Other Materials	Corrosion protection, Insulation, Coating & Painting							

Source: World Bank data.

Figure 4—G Supply Chain (Zainaddin, 2022)

Figure 4 categorizes and prioritizes each implementation item. Key high-priority implementation items include strategy and structure, where the organization retains its hybrid decentralized model; however, the corporate team must enhance its role within the company. This team is tasked with becoming the authoritative body on related strategies and policies and plans to foster alignment by improving active communication across all business units. In terms of process and governance, while the organization has well-documented procedures and policies, there is a lack of alignment in their implementation. To address this, the organization has established monthly meetings and follow-up sessions aimed at enhancing the dissemination and adherence to these processes. Regarding systems and tools, the organization demonstrates substantial usage of software and tools that support its operations, with each business unit accessing a unified information technology (IT) platform for managing metrics that feed into board reporting. The emphasis remains on continuing to integrate these systems while ensuring data accuracy. Lastly, managing the change required to implement specific HSE&SR strategies necessitates the development of a detailed management-of-change document, which will be regularly updated to cater to evolving demands. A timeline has been created to facilitate the implementation of these key focus items, focusing on both short-term and long-term objectives.



Figure 5—Organization of implementation items

Case Studies Around the Globe

Nigeria's local content policy in the oil and gas sector aims to boost local participation, retain capital, and create jobs by prioritizing Nigerian contractors. The industry spends over \$15 billion annually on services like fabrication and engineering, presenting a significant opportunity for local companies. However, challenges persist regarding the quality of service, expertise, and global competitiveness of these companies. The policy's success depends on local companies growing technically, financially, and managerially to meet international standards. Contracts should incentivize timely, high-quality delivery, while profits should be reinvested in research and development (R&D). The Nigerian Content Act favors Nigerian operators and indigenous companies with adequate resources. R&D is crucial for value creation, encouraging partnerships with universities and academics. Cooperative project execution, clear project understanding, technical support, project management, and performance-based contracts are vital for enhancing value delivery and ensuring the policy's success (Idachaba and Wokoma, 2014).

A North American safety provider implemented an accelerated workforce solution for stringent 60% local content requirements and a tight deadline to provide H₂S and CO₂ safety services for ten offshore rigs in Latin America. The organization executed a rapid nationalization program that involved intensive local recruitment, training exceeding regulatory standards, and customization of the corporate quality management system. Through on-the-job mentoring, local hires transitioned into onsite technician roles within the first month that achieving 75% local content and surpassing initial targets. Within twelve months, 100% nationalization was attained while establishing a scalable workforce model for regional expansion (MacDonald et al., 2015).

Strategies for improving the value delivery are key strategies that can be used to improve the quality of the delivery of these local companies is a cooperative execution of projects. This strategy was adopted in the execution of the Manifa project, where a drastic change in the financial projections made the project uneconomical following the original concepts. The management of Saudi Aramco worked together with the company to redesign and renegotiate the contracts and this resulted in the execution of the project in a

manner that increased the capacity of the contractors and resulted in a 2 billion dollar cost saving with the project being delivered 2 years ahead of schedule (Henni, 2013).

Case Study: Weatherford Saudi Arabia

Weatherford Saudi Arabia is a leading provider of oilfield services and equipment, operating within the Kingdom's dynamic and strategically important oil and gas sector. With a long-standing presence in Saudi Arabia, Weatherford has established itself as a key partner in the development and optimization of the Kingdom's OIL and gas industry. The company's operations span a wide range of activities, including drilling, evaluation, completion, production, and intervention services. Weatherford Saudi Arabia has demonstrated a strong commitment to local content development, aligning its business practices with the Kingdom's Vision 2030 objectives. The company recognizes that local content is not only a strategic imperative for the Saudi economy but also a key driver of its own long-term success through prioritizing local sourcing, workforce development, and technology transfer. Weatherford aims to contribute to the growth and diversification of the Saudi economy while enhancing its own competitiveness and sustainability.

Weatherford Saudi Arabia has implemented a range of local content initiatives across its operations. These initiatives encompass various aspects of the company's business, including procurement, human resources, and technology. In the area of procurement, Weatherford actively seeks to source goods and services from local suppliers whenever possible. The company has established a dedicated team responsible for identifying and evaluating local suppliers, providing them with technical assistance, and integrating them into Weatherford's supply chain.

In the area of human resources, Weatherford has implemented comprehensive workforce development programs aimed at attracting, training, and retaining Saudi nationals. These programs include scholarships for Saudi students, internships for recent graduates, and on-the-job training opportunities for experienced professionals. The company also provides leadership development programs to prepare Saudi employees for management positions.

Weatherford's commitment to local content has yielded significant benefits for its business in Saudi Arabia. By prioritizing local sourcing, the company has reduced its reliance on foreign imports, lowered its transportation costs, and shortened its lead times. By investing in workforce development, Weatherford has built a highly skilled and motivated workforce, reducing its dependence on expatriate labor. By engaging in technology transfer, the company has enhanced its reputation as a trusted partner and strengthened its relationships with local stakeholders.

However, Weatherford has also encountered challenges in implementing its local content programs. One of the main challenges is the limited availability of qualified local suppliers in certain areas. The company has addressed this challenge by providing technical assistance to local suppliers, helping them to improve their quality standards and meet Weatherford's requirements. Another challenge is the high cost of training and developing Saudi nationals. Weatherford has addressed this challenge by partnering with local universities and vocational schools to provide cost-effective training programs.

Weatherford's Five-Pillar Framework for KSA Local Content Roadmap

Weatherford's strategic local content enhancement plan represents a multi-faceted commitment to aligning with Saudi Arabia's vision 2030 objectives, focusing on technology sovereignty, economic diversification, and sustainable talent development. This integrated framework targets five core pillars as R&D localization, advanced manufacturing, supplier ecosystem development, national talent cultivation, and export enablement that are synergized through a unified operational ecosystem.

R&D Center KSA: Bridging Critical Technology Gaps

A fundamental pillar of Weatherford's strategic framework for enhancing in-country value (ICV) within the Kingdom of Saudi Arabia's oil and gas sector is a concentrated emphasis on innovative research and development (R&D) scope. The core objective is to engineer and validate cutting-edge technologies specifically designed to mitigate endemic operational challenges encountered in the Kingdom's unique geological and production environments via strategically investing in localized R&D infrastructure and expertise, Weatherford directly addresses pre-existing technological voids and simultaneously empowers the national engineering and research workforce with advanced tools, training, and developmental frameworks requisite for driving innovation. This proactive initiative is poised to cultivate indigenous technical talent and strategically position Saudi Arabia as a burgeoning nucleus for technological advancements critical to the evolution of the oil and gas domain. The localization of these R&D endeavors is crucial for optimizing Weatherford's operational efficacy and augmenting its capacity to respond with agility and precision to the distinct operational complexities prevalent in the region.

The R&D mandate encompasses a broad spectrum of technological development initiatives (Figure 6) that includes, but is not limited to, pioneering innovations in drilling and workover methodologies, casing exit technologies (Alharbi et al., 2025; Gamal et al., 2024b), advancements in managed pressure drilling (MPD) technologies (Whitfield, 2023), the conceptualization and deployment of smart completions systems, and the refinement of specialized well services solutions. A particular prominence is placed on the pervasive integration of digitalization and the application of artificial intelligence and machine learning models (Gamal et al., 2023) through advanced data analytics platforms designed to facilitate enhanced remote real-time monitoring of operational parameters, thereby optimizing operational excellence and minimizing non-productive time (Gamal, 2025; Gamal et al., 2024a).



Figure 6—R&D Cutting Edge Technologies

Furthermore, these strategic R&D initiatives are underpinned by a multifaceted approach that incorporates joint ventures with Saudi Aramco and local entities, the establishment of pertinent licensing agreements to facilitate technology transfer, and the implementation of comprehensive technical training programs designed to upskill the local workforce. Weatherford's sustained investment in these in-country R&D activities is intrinsically linked to its objective of co-developing bespoke, innovative solutions that directly address the specific technical and economic challenges, as well as capitalize on the unique opportunities characteristic of the Saudi Arabian oil and gas landscape.

SPARK Industrial Hub: Localized Manufacturing Excellence

Weatherford's commitment continues with the establishment of its manufacturing facility within the King Salman Energy Park (SPARK) project (Figure 7). This facility will focus on producing essential components, including screens, completion equipment, and liners, locally. The SPARK initiative represents a significant investment in localized manufacturing capabilities, which not only strengthens the supply chain but also generates local employment opportunities. Weatherford further contributes to the economic landscape by reducing import dependency, decreasing lead times for essential products, and enhancing overall project efficiency through opting for local manufacturing.



Figure 7—King Salman Energy Park (SPARK, 2025)

Local Supplier Database Development: Ecosystem Enablement

Another vital aspect of Weatherford's strategic plan is the development of a comprehensive supplier database and digital platform. This initiative seeks to identify and cultivate relationships with local vendors capable of providing high-quality goods and services. By prioritizing local suppliers, Weatherford aims to create a more resilient supply chain that can support ongoing operations while fostering economic growth in the local community. The establishment of this database will streamline procurement processes, ensuring that local businesses are engaged as pivotal partners in Weatherford's projects, thus enhancing the overall contribution to local content.

National Talent Management: 5-Year Development Pathway

Weatherford understands the significance of human capital in achieving its local content objectives. To this end, the company has implemented a robust talent management strategy that includes a comprehensive five-year action plan for career development. This plan focuses on training initiatives designed to equip Saudi nationals with the skills and competencies required in the oil and gas sector. Weatherford aims to foster a skilled workforce that aligns with the aspirations of Vision 2030 by emphasizing local talent management, ultimately contributing to job creation and economic diversification. The commitment to developing local human resources also paves the way for sustainable national employment.

Export Enablement via SPARK: Regional Hub Strategy

The SPARK manufacturing hub will serve as a springboard for regional export growth, transforming Saudi Arabia into a technology re-export center. Initial product focus covers customized sand screens for challenging operations, intelligent completion systems with integrated monitoring, and digital well integrity solutions. Target markets include Gulf Cooperation Council (GCC) neighbors and other countries through leveraging Saudi Arabia's trade agreements under the GCC common market, and African continental free trade area, and global markets. This strategic focus on exports not only diversifies income streams but also positions Saudi Arabia as a competitive player on the international stage in oil and gas technology and services. This export-driven mindset aligns with the Kingdom's broader vision of becoming a leading hub for industrial activities in the region, thereby maximizing the economic benefits derived from local investments.

Integrated Ecosystem Implementation

The integrated ecosystem implementation, as shown in Figure 8, is conceptualized as a comprehensive framework that synergizes multiple components to enhance local content and drive economic growth in the Kingdom of Saudi Arabia. At the core of this ecosystem is the talent program, which emphasizes developing a skilled workforce. This program is crucial because it feeds directly into the engineering teams along with R&D KSA, fostering a pipeline of talent that can effectively contribute to technological advancements. The workforce developed through the talent program facilitates tech transfer, which is essential for bringing innovative practices and technologies to local operations. SPARK manufacturing serves as the backbone of local production capabilities. Here, local sourcing of raw materials is prioritized, ensuring that materials used in manufacturing processes are obtained from domestic suppliers, which enhances the local supplier database. This database is vital for streamlining procurement and supporting local businesses.

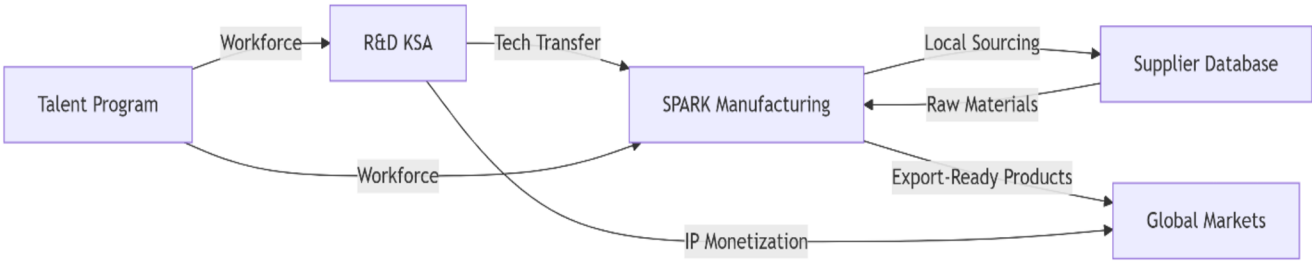


Figure 8—Weatherford KSA Integrated Ecosystem Implementation

Furthermore, SPARK Manufacturing is designed to produce export-ready products, allowing for broader market access. The output from this manufacturing process doesn't just cater to local needs but also positions Saudi Arabia to tap into global markets. This makes the entire ecosystem self-sustaining and dynamic, enhancing the region's competitiveness.

Besides, the concept of monetization intellectual property (IP) that plays a significant role in the integrated ecosystem through developing intellectual property within Saudi Arabia and leveraging it effectively, Weatherford aims to ensure that the benefits of local innovations are maximized, further supporting economic growth and establishing the Kingdom as a hub for O&G technology. The Integrated ecosystem implementation encapsulates a holistic approach that connects workforce development, R&D, local manufacturing, and global market strategies, driving sustainable growth and enhancing local value creation.

Accomplishments and Future Directions

The IKTVA program marks a significant milestone, celebrating a decade of localization progress. As of 2024, 67% of Saudi Aramco's procurement spending contributes directly to local content, underscoring the program's effectiveness in fostering local economic engagement. Since its inception, IKTVA has made an estimated cumulative contribution of \$240 billion to the Kingdom's gross domestic product (GDP). The program has also played a crucial role in job creation, with approximately 200,000 direct and indirect jobs sustained as a result of its initiatives since 2015. Furthermore, IKTVA has facilitated the establishment of 350 new manufacturing facilities, contributing to the diversification of the local economy. Notably, it has enabled the production of 47 new products that had not been manufactured in Saudi Arabia prior to the program's launch. Additionally, companies participating in the IKTVA initiative have demonstrated robust growth in research and development capabilities, with an average annual growth rate of 33% in R&D spending. These results highlight the substantial impact of IKTVA in driving localization and economic development within the Kingdom (IKTVA, 2025).

Table 1—Iktva 10 Years Of Localization Progress (IKTVA, 2025)

Metric	Details
67%	Percentage of Aramco's procurement spending contributing to local content as of 2024.
\$240 Billion	Estimated cumulative GDP contribution to the Kingdom since inception.
200,000	Estimated number of direct and indirect jobs created and sustained by the IKTVA program since 2015.
350	Number of new manufacturing facilities enabled by IKTVA since its launch.
47	Number of new products manufactured for the first time in Saudi Arabia thanks to IKTVA.
33%	Average annual growth rate in R&D spending among IKTVA companies.

In conclusion, Weatherford's strategic plan for local content enhancement is multifaceted and aligned with the Kingdom of Saudi Arabia's vision 2030. Through strategic R&D, local manufacturing through the SPARK initiative, establishing a comprehensive supplier platform, and developing local talent, Weatherford is set to play a pivotal role in advancing local content in the oil and gas sector. Through these initiatives, Weatherford not only aims to boost its operational efficiencies but also contribute to broader economic goals that promise sustainable growth and job creation in the Kingdom. This strategic approach demonstrates Weatherford's commitment to being a responsible partner in Saudi Arabia's progress, fostering a collaborative environment that benefits local communities and strengthens the nation's economy. The combined efforts will drive long-term benefits, enabling the Kingdom to optimize its resources and position itself as a leader in the global oil and gas industry.

In conclusion, the journey of local content enhancement in Saudi Arabia is a dynamic process with significant achievements and future potential. Through learning from past experiences, addressing current challenges, and strategically focusing on innovation, talent development, and collaborative ecosystems, the Kingdom is well-positioned to maximize the value derived from its resources and build a diversified and resilient economy for the future.

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References

1. Acheampong, T., Ashong, M., Svanikier, V.C., 2016. An assessment of local-content policies in oil and gas producing countries. *The Journal of World Energy Law & Business* **9**, 282–302. <https://doi.org/10.1093/jwelb/jww019>
2. Alayed, S., Almushaiti, M., Bin Helayel, H., Alnasser, A., 2025. Saudi Arabia's Business Transformation: Strategies for Success in a Changing Economy - Scholar Publishing. URL <https://scholarpublishing.org/sse/eb362/> (accessed 6.1.25).
3. Alharbi, A., Petre, R., Gamal, H., 2025. Casing Exit Excellence: a Holistic Strategy of Integrated Engineering Models and Well Integrity Analytics for Extended Reach Wells.
4. d'Almeida, A.L., Apolinário, B. de P., Ribeiro, L.F., 2019. Brazilian Local Content Policy: An Overview and Analysis of Adjustment Clauses Applications.
5. Denney, D., 2011. Managing Use of Local Content in Upstream Oil and Gas. *Journal of Petroleum Technology* **63**, 95–96. <https://doi.org/10.2118/0811-0095-JPT>
6. Deringer, H., Erixon, F., Lamprecht, P., 2018. The Economic Impact of Local Content Requirements: A Case Study of Heavy Vehicles.

7. Emielu, E., 2005. A Local Content Knowledge Management System as a Strategy for Achieving Sustainable Local Content Development in the Nigerian Oil & Gas Industry.
8. Gamal, H., 2025. Technological Integration and Innovative Strategies Harnessing Artificial Intelligence for Operational Excellence [WWW Document]. <https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/979-8-3693-6447-5.ch009>. URL <https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/359302> (accessed 5.29.25).
9. Gamal, H., Albahri, S., Essa, H., Srivastava, M., Alghamdi, N., Shaikh, W., 2024a. Driving Operational Excellence: Cost Savings and Performance Optimization Through Tripping and Connection Time Optimization.
10. Gamal, H., Elkatatny, S., Al Gharbi, S., 2023. Rig Sensor Data for AI-ML Technology-Based Solutions: Research, Development, and Innovations. *Presented at the ADIPEC*, OnePetro. <https://doi.org/10.2118/216429-MS>
11. Gamal, H., Gomez, H., Alshalan, M., Alshaarawi, A., 2024b. Successful Field Trials for Tracking Casing Exit: Innovative Sensing Technology Application, in: Day 2 Tue, February 13, 2024. Presented at the International Petroleum Technology Conference, IPTC, Dhahran, Saudi Arabia, p. D021S061R002. <https://doi.org/10.2523/IPTC-23750-MS>
12. Hammann, E., 2016. Three Years Experience in Engaging Local Content at Project Planning Phase.
13. Henni, A., 2013. Huge Saudi Field Begins Production Ahead of Schedule [WWW Document]. *JPT*. URL <https://jpt.spe.org/huge-saudi-field-begins-production-ahead-schedule> (accessed 6.2.25).
14. Idachaba, F.E., Wokoma, E., 2014. Project Management, Local Content and Value Delivery, in: Offshore Technology Conference. *Presented at the Offshore Technology Conference*, OTC, Houston, Texas, p. D041S051R005. <https://doi.org/10.4043/25227-MS>
15. IKTVA, 2025. *IKTVA* [WWW Document]. URL <https://www.aramco.com/en/what-we-do/commercial-ecosystems/iktva> (accessed 6.3.25).
16. MacDonald, S., Shah, B., Almeida, C., Daher, E., 2015. Fulfilling Local Content Requirements Within The First Six Months of Country Operation: A Case Study.
17. Seregin, N., 2014. Local Content Challenges in Upstream Overseas Projects.
18. SPARK, 2025. *SPARK* [WWW Document]. URL <https://www.spark.sa/> (accessed 6.4.25).
19. Whitfield, S., 2023. Weatherford MPD software delivers time savings by optimizing pressure management. Drilling Contractor. URL <https://drillingcontractor.org/weatherford-mpd-software-delivers-time-savings-by-optimizing-pressure-management-66823> (accessed 6.4.25).
20. Zainaddin, I., 2022. Enhancing Operational Business Decision Making by Better Understanding Gcc's Oil and Gas Energy Sector Local Content Programs.